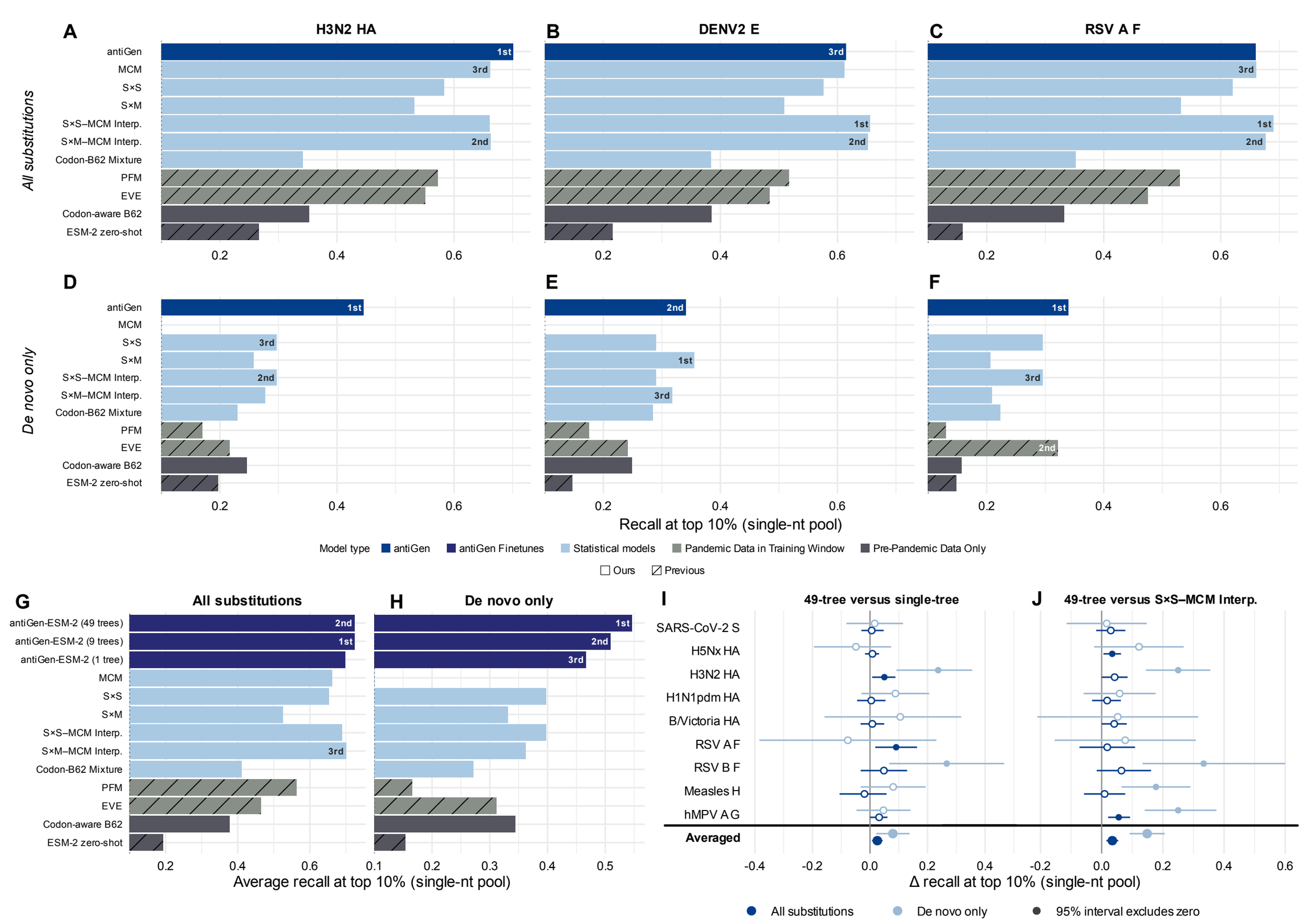

A
H3N2 HA
B
DENV2 E
C
RSV A F
All substitutions
De novo only
antiGen
MCM
S×S
S×M
S×S–MCM Interp.
S×M–MCM Interp.
Codon-B62 Mixture
PFM
EVE
Codon-aware B62
ESM-2 zero-shot
1st
2nd
3rd
0.2
0.4
0.6
D
E
F
Recall at top 10% (single-nt pool)
Model type
antiGen
antiGen Finetunes
Statistical models
Pandemic Data in Training Window
Pre-Pandemic Data Only
Ours
Previous
G
All substitutions
H
De novo only
antiGen-ESM-2 (49 trees)
antiGen-ESM-2 (9 trees)
antiGen-ESM-2 (1 tree)
Average recall at top 10% (single-nt pool)
I
49-tree versus single-tree
J
49-tree versus S×S–MCM Interp.
SARS-CoV-2 S
H5Nx HA
H3N2 HA
H1N1pdm HA
B/Victoria HA
RSV A F
RSV B F
Measles H
hMPV A G
Averaged
-0.4
-0.2
0.0
0.2
0.4
0.6
Δ recall at top 10% (single-nt pool)
All substitutions
De novo only
95% interval excludes zero